NewsImages in MediaEval 2026 – Automated Image Recommendations with Retrieval and Generation Techniques for News Articles Thumbnails

Lucien Heitz[1,*], Bruno N. Sotic[2], Ali A. Katamjani[3], Qi Bi[3], Bram Bakker[4], Luca Rossetto[5] and Jaap Kamps[2]

[1]*University of Zurich, Zurich, Switzerland*
[2]*University of Amsterdam, Amsterdam, the Netherlands*
[3]*Utrecht University, Utrecht, the Netherlands*
[4]*Radboud University Nijmegen, Amsterdam, the Netherlands*
[5]*Dublin City University, Dublin, Ireland*

Abstract
The NewsImages challenge in MediaEval 2026 focuses on matching news articles with appropriate thumbnails. Participants receive a large set of English-language articles from international publishers. Given the text of a news article, the goal of the challenge is to recommend a fitting image by creating automated workflows for image retrieval and/or generation. The overall quality of the image recommendation is evaluated in a crowd-sourced online event where image fit and relevance are assessed. This task overview paper provides details on the challenge, the technical aspects of the run submissions, and the online user study for the evaluation. We includes a summary of the most important lessons learned from past NewsImages challenge iterations.

1. Introduction

Thumbnails for news articles are an important factor in capturing the reader's attention on online news platforms [1]. Newsrooms can use a wide range of image retrieval or image generation approaches to create fitting visuals (cf. [2, 3]). Recent approaches to automated image retrieval and generation have been shown to compete with, and in some cases even outperform, editorial selection of visuals [4]. This year's challenge task is a seamless continuation of the 2025 edition [5], in which we systematically explore the capabilities of image generation techniques and compare their recommendations with editorial selection for news thumbnails. The goal of NewsImages 2026 is to compare automated and editorial approaches across different news topics and to assess their performance in an online survey. This helps us identify the advantages and disadvantages to understand the implications for journalism and news personalization. NewsImages 2026 places great emphasis on the user perspective. Therefore, we are not only interested in technical papers and contributions to image retrieval and generation, but also in qualitative work on user perception or trust.[1]

MediaEval'26: Multimedia Evaluation Workshop, June 15–16, 2026, Amsterdam, Netherlands and Online
heitz@ifi.uzh.ch (L. Heitz); nadalic.sotic@uva.nl (B. N. Sotic); a.ahmadikatamjani@uu.nl (A. A. Katamjani); q.bi@uu.nl (Q. Bi); bram.bakker2@ru.nl (B. Bakker); luca.rossetto@dcu.ie (L. Rossetto); kamps@uva.nl (J. Kamps)
0000-0001-7987-8446 (L. Heitz); 0009-0003-2122-2235 (B. N. Sotic); 0009-0008-9545-2814 (A. A. Katamjani); 0000-0002-1047-4790 (Q. Bi); 0009-0009-8615-1056 (B. Bakker); 0000-0002-5389-9465 (L. Rossetto); 0000-0002-6614-0087 (J. Kamps)

CEUR Workshop Proceedings (CEUR-WS.org)
[1]See *Quest for Insight* for more topics: https://multimediaeval.github.io/editions/2026/tasks/newsimages

2. Task Description

NewsImages 2026 explores matching news articles with automated image recommendations. To that end, participants receive a large set of English-language articles from international news organizations and media outlets. Given the text of a news article, the goal of this task is to build a recommendation pipeline that uses retrieval techniques, generative AI, or a combination of both to output a suitable image for an article thumbnail. We briefly outline the available resources, the dataset, and the final submission. The complementary online README file provides more information and technical details.[2]

Resource Selection Staying true to MediaEval's principles of promoting reproducible research, we ask participants to share their entire image retrieval pipelines and generation workflows (including all parameters and links to models to reproduce the image recommendations). The image retrieval/generation pipeline *must not* rely on any closed-source APIs or resources that are not publicly available. For image retrieval, we recommend public datasets like *Yahoo-Flickr Creative Commons 100 Million* (YFCC100M)[3] or *Public Domain 12 Million* (PD12M).[4] For image generation, we require the models to be open source and/or open weights.

Training Dataset In terms of the training dataset, participants are given a collection of $8,500$ news articles with images (the article text is in English, collected during 2022–2023 by GDELT).[5] Each new item includes the following properties: $article\_id$ (internal ID of news article used for submission), $article\_url$ (URL of the original news article), $article\_title$ (title of the news article, may include lead), and $image\_id$ (internal ID of original image for which a copy is shared). The article text itself is not shared, but participating groups are free to retrieve it from the original source. The dataset is the same as in NewsImages 2025. The main idea behind reusing it for training purposes is that it allows participating teams to build upon existing workflows and compare their recommendations with past results.

Test and Evaluation Dataset There will be a separate test and evaluation dataset consisting of $15,000$ articles. These articles and images are sourced from MIND.[6] The format will be the same as the training dataset. Complementing this, there will be a small selection of 100 historical articles. These are headlines and scans from 19^{th}-20^{th}-century newspaper articles.[7] All articles are in English. Some news headlines from historical articles are machine-translated from Dutch to English. The online survey contains 35 random articles from the test dataset.

Run Submissions Participating teams are encouraged to explore different image styles (e.g., photo-realistic or caricature), image subjects (e.g., people, places, things), and combinations thereof. Teams can submit multiple runs. However, all runs must be substantially different from one another, must not include any original article images shared in the task datasets, and they must be properly documented by each team.[8]

---

[2] README available on GitHub: https://github.com/Informfully/Challenges
[3] YFCC100M website: https://multimediacommons.org
[4] PD12M on Huggingface:https://huggingface.co/datasets/Spawning/PD12M
[5] GDELT website: https://gdeltproject.org
[6] MIND website: https://msnews.github.io
[7] Please see separate *Quest for Insight* publication for details on the historical news images [6].
[8] For instructions on how to write the accompanying Working Notes Paper, please go to the official MediaEval 2026 website: https://multimediaeval.github.io/editions/2026

3. Experiment Guidelines

The NewsImage iteration of 2025 was the first to systematically compare image retrieval and generation pipelines. Based on these results, we have several key insights for future submissions that we would like to share with participants regarding the impact of state-of-the-art models, building hybrid workflows, exploring different style options, and being mindful of ethical issues associated with automated selection/retrieval of visuals.[9]

State-of-the-Art Models For image retrieval, variants of CLIP [2] have been among the best-performing models in the past (e.g., OpenCLIP submissions [7]). For image generation, please consider a top-ranking open-weight text-to-image model[10] or text-to-video model.[11]. They have been among the most successful approaches in the prior iteration of the task [8, 9]. These models can be used standalone or combined with different preprocessing steps. For example, prompting can be augmented by additional text-to-text models to get richer input prompts for retrieval [10, 11, 12, 13] and generation [8, 14]. You can think of, e.g., creating specific prompting templates for different article categories or keywords within the title [15, 16].

Hybrid Workflows Focus on hybrid pipelines that combine the strengths of image retrieval and image generation (e.g., retrieving images when a specific person is mentioned, and generating images when you need a generic picture of a place, cf. [17, 16]. This requires building article-specific workflows that you can switch between, e.g., based on article category [15]. Another approach is to use retrieval and generation workflows in parallel to create multiple image recommendations. Once there is a sufficiently large pool of candidate images, it is possible to add a post-processing step where another model scores [11], reranks [12], or filters out images [18] to determine the best candidate, or to create a collage thereof [19]. This can be done either once or iteratively, until a certain quality threshold is reached.

Style Preferences Art style and image subjects have a significant impact on performance Previous work showed that cartoons and caricatures were generally well-received and achieved a better-than-average score [14, 8]. Furthermore, they provide a reliable way to generate images that also comply with existing media policies against creating realistic images, as they could help spread misinformation and/or fake news. We recommend conducting pretests with human judges to assess the performance of the runs. Offline metrics, such as text-image similarity or accuracy, were shown to be a poor indicator of final performance in online evaluation [12, 18].

Ethical Guidelines There is a wide range of ethical issues with the automated selection and/or generation of visuals for news [20]. Keep in mind the social responsibility of editors [21, 22] and that news is an integral part of democratic societies [23, 24]. Therefore, we highly recommend that participants adhere to editorial standards and guidelines for image generation. We are particularly interested in non-photorealistic images that do not suggest they accurately represent real events, to avoid misleading or deceiving readers.[12]

---

[9] All workflows and image recommendations of NewsImages 2025 are available on GitHub for teams to explore and benchmark their runs: https://github.com/Informfully/Challenges/tree/main/newsimages25

[10] Online text-to-image leaderboard: https://artificialanalysis.ai/image/leaderboard/text-to-image?open-weights=true

[11] Online text-to-video leaderboard: https://artificialanalysis.ai/text-to-video/arena?tab=leaderboard-text

[12] For details on the guidelines considered for the subtasks, we refer to the AI-CODE Deliverable D3.1 on user-centered requirements definition (AI-CODE Consortium, 2025).

4. Evaluation

The image recommendations of the submitted runs will be evaluated in a crowd-sourced online event. All participating teams, task organizers, and an external jury will take part in the event. Image fit is the primary criterion for evaluating and assessing images. Fit focuses on whether the images accurately capture the article's key attributes, without depicting any elements not present in the article. During the event, participants are presented with a news headline and a list of image recommendations. The fit of each image is rated on a 5-point Likert scale. The winning team is determined by the highest average image-fit rating achieved by their best submitted run. There will be a baseline using the editors' original images, as well as a state-of-the-art text-to-image baseline presented by the task organizers.

5. Conclusion

NewsImages 2026 builds on the 2025 iteration, which first combined image retrieval and generation techniques to produce thumbnail recommendations based on news headlines. With this continuation, our goal is to leverage techniques and insights from previous submissions to create new automated workflows. Furthermore, we want to reiterate that the NewsImages challenge focuses on both the technological aspects and the user's perspective. We also look for contributions that assess the impact of recommendations on audiences and/or critically reflect on the requirements, risks, and opportunities presented by these new techniques. In doing so, we encourage a broad range of research fields and disciplines to participate in the challenge and share their results and insights for generating new image recommendations.

Acknowledgments

Ali A. Katamjani, Bram Bakker, Qi Bi, Bruno N. Sotic, and Jaap Kamps are partly funded by the HAICu project from the Netherlands Organization for Scientific Research (NWO NWA #1518.22.105). Jaap Kamps is further supported by the University of Amsterdam (AI4FinTech program) and ICAI (AI for Open Government Lab). Views expressed in this paper are not necessarily shared or endorsed by those funding the research.

Declaration on Generative AI

During the preparation of this task overview paper, the authors used Grammarly for grammar and spelling checks. After using this service, the authors reviewed and edited the text as needed. They take full responsibility for the publication's content.

References

[1] T. Thomson, To see and be seen: The environments, interactions and identities behind news images, Rowman & Littlefield, 2019.

[2] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, in: M. Meila, T. Zhang (Eds.), Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event, volume 139 of *Proceedings of Machine Learning Research*, PMLR, 2021, pp. 8748–8763.

[3] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022, IEEE, 2022, pp. 10674–10685.
[4] L. Heitz, A. Bernstein, L. Rossetto, An empirical exploration of perceived similarity between news article texts and images, in: Working Notes Proceedings of the MediaEval 2023 Workshop, 2024. URL: http://ceur-ws.org/Vol-3658/paper8.pdf.
[5] L. Heitz, L. Rossetto, B. Kille, A. Lommatzsch, M. Elahi, D.-T. Dang-Nguyen, Newsimages in mediaeval 2025 – comparing image retrieval and generation for news articles, in: Working Notes Proceedings of the MediaEval 2025 Workshop, 2025. URL: https://2025.multimediaeval.com/paper1.pdf.
[6] L. Heitz, B. N. Sotic, Q. Bi, J. Kamps, Assessing the viability of image-to-image models for restoring historical photographs of news articles, in: Working Notes Proceedings of the MediaEval 2026 Workshop, 2026. URL: https://2026.multimediaeval.com/paper8.pdf.
[7] L. Heitz, Y. K. Chan, H. Li, K. Zeng, A. Bernstein, L. Rossetto, Prompt-based alignment of headlines and images using openclip, in: Working Notes Proceedings of the MediaEval 2023 Workshop, 2024. URL: http://ceur-ws.org/Vol-3658/paper7.pdf.
[8] M. Khan, A. Subhani, M. Rafi, A. Tahir, Balancing relevance and compliance: Text-to-image methods for news articles visualization (2025). URL: https://2025.multimediaeval.com/paper34.pdf.
[9] D. S. Madhavan, L. P. S. Rao, K. K. Subramani, J. Govindaraj, A. Elanchezhiyan, A. R. Thavasi, Dual-pipeline multimedia enrichment for news articles via image generation and retrieval (2025). URL: https://2025.multimediaeval.com/paper26.pdf.
[10] H. Dillibabu, N. K. Kandasamy, A. S. Rajashree, A comparative study of vision-language models for news image-headline matching (2025). URL: https://2025.multimediaeval.com/paper52.pdf.
[11] D. Galanopoulos, A. Goulas, V. Mezaris, Cross-modal image recommendation for news articles by multimodal foundation models-based retrieval-reranking (2025). URL: https://2025.multimediaeval.com/paper11.pdf.
[12] B. Khaertdinov, A. Ganesh, M. Popa, N. Tintarev, Beyond similarity: Two-stage retrieval for news image search (2025). URL: https://2025.multimediaeval.com/paper20.pdf.
[13] R. Swaminathan, S. Thangapandiyan, M. Palaniappan, S. Esakki, Clip-based news image retrieval with keyword-enriched prompts for mediaeval 2025 newsimages task (2025). URL: https://2025.multimediaeval.com/paper21.pdf.
[14] T.-K. Nguyen, H.-D. Nguyen, M.-T. Tran, Less is more: A dual-filtering perspective on prompt design for news thumbnail generation (2025). URL: https://2025.multimediaeval.com/paper40.pdf.
[15] B. Bakker, X. Wang, Improving text-to-image retrieval of news articles, class-aware fine-tuning of clip and the use of lead text (2025). URL: https://2025.multimediaeval.com/paper16.pdf.
[16] X. Wang, B. Bakker, Diffusion-based approaches for newsimage generation: A comparative study of sdxl variants (2025). URL: https://2025.multimediaeval.com/paper17.pdf.
[17] P. Balasundaram, H. Saminathan, B. Arivalagan, S. Subramaniam, Newsimages: Clip–faiss retrieval and diffusion-based generation for editorial news thumbnails (2025). URL: https://2025.multimediaeval.com/paper46.pdf.
[18] L. P. S. Rao, D. S. Madhavan, S. Thota, T. Ravialagappan, V. Muralidharan, S. K. Shivaanee, A systematic evaluation of vision-language models for news image retrieval: A two-stage retrieve-and-rerank pipeline analysis (2025). URL: https://2025.multimediaeval.com/paper43.pdf.
[19] M. Palaniappan, Y. P. Harikrishnan, H. Jayakumar, K. M. Kalyani, Newsimages: Faiss-based image retrieval and stable diffusion generation for diverse news articles (2025). URL: https://2025.multimediaeval.com/paper12.pdf.
[20] T. Thomson, R. J. Thomas, P. Matich, Generative visual ai in news organizations: Challenges, opportunities, perceptions, and policies, Digital Journalism 13 (2025) 1693–1714.
[21] N. Helberger, On the democratic role of news recommenders, Digital Journalism 7 (2019) 993–1012.
[22] A. Bernstein, C. De Vreese, N. Helberger, W. Schulz, K. Zweig, L. Heitz, S. Tolmeijer, et al., Diversity in news recommendation, Dagstuhl Manifestos 9 (2021) 43–61.
[23] L. Heitz, K. Rozgonyi, B. Kostic, D. Wagner, J. Haas, AI in Content Curation and Media Pluralism, Organization for Security and Co-operation in Europe, Austria, 2021, pp. 56–70.
[24] L. Heitz, J. A. Lischka, A. Birrer, B. Paudel, S. Tolmeijer, L. Laugwitz, A. Bernstein, Benefits of diverse news recommendations for democracy: A user study, Digital Journalism 10 (2022) 1710–1730.